Representation theory and Toeplitz operators

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Abstract

A Toeplitz operator is the composition of a multiplication operator and an orthogonal projection. An interesting problem related to Toelitz operators is to find an abelian C*-algebra generated by Toeplitz operators. In the case of the Poincare disk $\mathbb{D}$ there are three families of symbols such that the C*-algebra generated by the Toeplitz operators with these symbols commutes. These families of symbols consist of those symbols that are invariant under the action fo certain subgroups of $\mathrm{Aut}(\mathbb{D})$. For the case of $\mathbb{B}^n$ we have a similar result: the Toeplitz operators that generate abelian C*-algebras are those whose symbols are invariant under the action of certain subgroups of $\mathrm{Aut}(\mathbb{B}^n)$. This behavior manifests itself in bounded symmetric domains; the proof of existence and the selection of subgroups rely essentially on techniques from representation theory. The criterion is that if the holomorphic discrete series, when restricted to the subgroup H, yields a multiplicity-free representation, ten the invariant symbols under the action of H will cause the Toeplitz operators to generate a commutative C* algebra.

The Existence Problem of Commutative C^*-Algebras on Bergman Spaces

Let D be a bounded symmetric domain. The weighted Bergman space $\mathcal{H}^2_\lambda(D)$ is the closed subspace of $L^2(D,\mu_\lambda)$ consisting of holomorphic functions.

For a given symbol $\varphi \in L^\infty(D)$, the **Toeplitz operator** $T^{(\lambda)}_\varphi : \mathcal{H}^2_\lambda(D) \to \mathcal{H}^2_\lambda(D)$ is defined as the multiplication operator M_φ followed by the Bergman projection B_λ:

$$T^{(\lambda)}_\varphi = B_\lambda \circ M_\varphi$$

In general, Toeplitz operators do not commute ($T^{(\lambda)}_\varphi T^{(\lambda)}_\psi \neq T^{(\lambda)}_\psi T^{(\lambda)}_\varphi$). An interesting problem is to find an abelian C^*-algebra generated by Toeplitz operators. We can approach this problem by studying the properties of the symbols—that is, by considering the algebra generated by Toeplitz operators whose symbols possess a specific property, such that the resulting algebra is abelian. This approach has the advantage of allowing us to study functions rather than operators.

The Unit Ball $\mathbb{B}^n = \mathrm{SU}(n,1)/\mathrm{S}(\mathrm{U}(n)\times\mathrm{U}(1))$ case

Initially, the problem for $\mathbb{D}$ was approached using tools from hyperbolic geometry; however, it was possible to generalize it to the case of the unit ball $\mathbb{B}^n$ by re-expressing the geometric properties in terms of group actions. When H is a maximal abelian subgroup, the H-invariant essentially bounded symbols result in the C^*-algebra generated by these operators being abelian.

Representation theory

A representation of a locally compact group G over a Hilbert space $\mathcal{H}$ is just a topological group homomorphism

$$\pi : G \to U(\mathcal{H})$$

where $U(\mathcal{H})$ is endowed with the strong operator topology.

Holomorphic discrete series: The bridge

Let us consider the bounded symmetric domain $D = G/K$ and the weighted Bergman space $\mathcal{H}^2_\lambda(D)$ for $\lambda > p-1$. The holomorphic discrete series is a representation of the universal covering group $\tilde{G}$. The group action is given by:

$$\pi_\lambda(g)f(z) = j(g^{-1},z)^{\frac{\lambda}{p}} f(g^{-1}z)$$

where $j(g,z)$ is the complex Jacobian.

Intertwining Operators and Commutativity

The connection between Toeplitz operators and representation theory is established by the following equivalence [1]:

Lemma (Intertwining Condition)

Let $\tilde{H}$ be a subgroup of $\tilde{G}$ and $\varphi \in L^\infty(D)$. The Toeplitz operator $T^{(\lambda)}_\varphi$ intertwines the restriction $\pi_\lambda|_{\tilde{H}}$ **if and only if** its symbol φ is $\tilde{H}$-invariant.

Main Theorem (Dawson, Ólafsson, Quiroga-Barranco, 2015)

Let H be a closed subgroup of G and $\mathcal{A}^H$ the subspace of H-invariant bounded symbols on D. If the algebra of intertwining operators $\mathrm{End}_{\tilde{H}}(\mathcal{H}^2_\lambda(D))$ is commutative, then the generated C^*-algebra $\mathcal{T}^{(\lambda)}(\mathcal{A}^H)$ is commutative.

In particular, this holds if H is a type I group and the restriction $\pi_\lambda|_{\tilde{H}}$ is **multiplicity-free**.

By applying this theorem to K, the maximal compact subgroup of G, we obtain a generalization of the $\mathbb{B}^n$ case.

Corollary

Let $\mathcal{A}^K$ be the set of bounded symbols that are K-invariant. Then, the C^*-algebra $\mathcal{T}^{(\lambda)}(\mathcal{A}^K)$ is commutative on the weighted Bergman space $\mathcal{H}^2_\lambda(D)$ for every $\lambda > p-1$.

Commutative subgroups of biholomorphisms of $\mathbb{B}^n$

There are exactly $n+3$ essentially different types of such subgroups [2]:

- **Quasi-elliptic:** Isomorphic to $\mathbb{T}^n$, acting on the unit ball $\mathbb{B}^n$.
- **Quasi-parabolic:** Isomorphic to $\mathbb{T}^{n-1}\times\mathbb{R}$, acting on the Siegel domain D_n.
- **Quasi-hyperbolic:** Isomorphic to $\mathbb{T}^{n-1}\times\mathbb{R}_+$, acting on D_n.
- **Nilpotent:** Isomorphic to $\mathbb{R}^{n-1}\times\mathbb{R}$, acting on D_n.
- **Quasi-nilpotent:** Isomorphic to $\mathbb{T}^k\times\mathbb{R}^{n-k-1}\times\mathbb{R}$ (for $0<k<n-1$), acting on D_n.
- **Invariant under the action of** $SO(n,1)\subset SU(n,1)$

References

[1] M. Dawson, G. Ólafsson, and R. Quiroga-Barranco, "Commuting Toeplitz operators on bounded symmetric domains and multiplicity-free restrictions of holomorphic discrete series," *Journal of Functional Analysis*, vol. 268, pp. 1711–1732, 2015.

[2] R. Quiroga-Barranco and N. Vasilevski, "Commutative C*-algebras of Toeplitz operators on the unit ball. I. Bargmann-type transforms and spectral representations of Toeplitz operators," *Integral Equations Operator Theory*, vol. 59, no. 3, pp. 379–419, 2007.

[3] R. Quiroga-Barranco and N. Vasilevski, "Commutative C*-algebras of Toeplitz operators on the unit ball. II. Geometry of the level sets of symbols," *Integral Equations Operator Theory*, vol. 60, no. 1, pp. 89–132, 2008.

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